Fall 2025 CPTS530 Homework 5
MATH 448-548
Numerical Analysis

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Abstract

This report presents solutions to selected problems from Homework 5, covering topics in matrix norms, iterative methods, and polynomial interpolation. Problem statements and theoretical foundations follow the course textbook [1] and lecture notes [2].

Contents

Problem 1 **2**

Problem 2 **3**

Problem 3 **4**

Problem 4 **5**

Problem 5 **6**

Problem 6 **7**

Problem 7 **8**

Problem 8 **9**

Problem 9 **10**

Appendix **11**

References **12**

Problem 1

Problem Statement (Textbook 4.4.1):

Show that the norms $\|\mathbf{x}\|_\infty$, $\|\mathbf{x}\|_2$, and $\|\mathbf{x}\|_1$ satisfy the postulates (1), (2), and (3) for norms.

Solution:

Problem 2

Problem Statement (Textbook 4.4.2):

Show that $\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1$ for all $\mathbf{x} \in \mathbb{R}^n$, and that equalities can occur, even for nonzero vectors.

Solution:

Problem 3

Problem Statement (Textbook 4.4.3):

(Continuation) Show that $\|\mathbf{x}\|_1 \leq n\|\mathbf{x}\|_\infty$ and $\|\mathbf{x}\|_2 \leq \sqrt{n}\|\mathbf{x}\|_\infty$ for $\mathbf{x} \in \mathbb{R}^n$.

Solution:

Problem 4

Problem Statement (Textbook 4.4.8):

Define

$$\|A\| = \sum_{i=1}^{n} \sum_{j=1}^{n} |a_{ij}|$$

Show that this is a matrix norm (that is, a norm on the linear space of all $n \times n$ matrices). Show that it is not subordinate to any vector norm. Does it conform to Equation (9) and Inequality (10)?

$$\|I\| = 1 \quad \text{(Equation 9)}$$

$$\|AB\| \leq \|A\|\|B\| \quad \text{(Inequality 10)}$$

Solution:

Problem 5

Problem Statement (Textbook 4.4.11):

Show that for the vector norm $\|\mathbf{x}\|_1$ defined in Equation (5), the subordinate matrix norm is

$$\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^{n} |a_{ij}|$$

where

$$\|\mathbf{x}\|_1 = \sum_{i=1}^{n} |x_i| \quad \text{(Equation 5)}$$

Solution:

Problem 6

Problem Statement (Textbook 4.4.33):

For any $n \times n$ matrix A, define

$$\|A\|_F = \left(\sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij}^2 \right)^{1/2}$$

(This is called the **Frobenius** norm.) Is this a subordinate matrix norm? Answer the same question for $\|A\| = \max_{1 \le i,j \le n} |a_{ij}|$. Prove that these equations define norms on the vector space of all $n \times n$ matrices.

Solution:

Problem 7

Problem Statement (Textbook 4.6.1b):

Program the Gauss-Seidel method and test it on the following system:

$$\begin{aligned} 3x + y + z &= 5 \\ 3x + y - 5z &= -1 \\ x + 3y - z &= 3 \end{aligned}$$

Analyze what happens when these systems are solved by simple Gaussian elimination without pivoting.

Solution:

Problem 8

Problem Statement (Textbook 6.1.1d):

Find the polynomial of least degree that interpolates these sets of data:

x	3	7	12
y	12	146	2

Solution:

Problem 9

Problem Statement (Textbook 6.1.8):

Discuss the problem of determining a polynomial of degree at most 2 for which $p(0)$, $p(1)$, and $p(\xi)$ are prescribed, ξ being any preassigned point.

Solution:

Appendix

References

[1] David Kincaid and Ward Cheney. *Numerical Analysis: Mathematics of Scientific Computing*. American Mathematical Society, Providence, RI, 3rd edition, 2009. Available from ProQuest Ebook Central. `https://ebookcentral.proquest.com/lib/wsu/detail.action?docID=4717637`.

[2] Alexander Panchenko. Math 448 lecture notes. Washington State University, Fall 2025.